



SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO4: Construct Context-Free Grammars, generate derivations and parse trees to demonstrate the syntactic structure of strings. (BL : 3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Duration : 1 Hour

CLASS TEST

Code of Conduct:

/ PERAM VIVEKANANDA REDDY(192372330)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Vivek

Criteria	Weight age	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total Marks (50)
Conceptual Understanding	30	3	3	3	3	3	15
Accuracy of Answer	25	2.5	2.5	2.5	2.5	2.5	12.5
Application of Knowledge	20	2	2	2	2	2	10
Completeness of Response	15	1.5	1.5	1.5	1.5	1.5	7.5
Clarity & Presentation	10	1	1	1	1	1	5
Total per Question	10 Marks	/10	/10	/10	/10	/10	/ 50



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1. Design a PDA to accept strings with 'n' occurrences of 'a' followed by 'n' occurrences of 'b', where 'n' is a non-negative integer.

i) $L = \{a^n b^n\}$

2. Convert the given context-free grammar (CFG) into a Pushdown Automaton (PDA) that recognizes the same language.

$$E \rightarrow E+E \mid E^*E \mid (E) \mid I$$

$$I \rightarrow Ia \mid Ib \mid I0 \mid I1 \mid a \mid b$$

3. Design a TM to accept the strings with 'n' occurrences of 'a' followed by 'n' occurrences of 'b', where 'n' is a non-negative integer

i) $L = \{0^n 1^n \mid n \geq 1\}$

4. Design a Pushdown Automata for the language representing a's followed by b's and the number of b's must be twice that of a's.

5. Construct a Turing Machine to recognize the language. Write the logic used for the design and the formal definition



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RUBRICS FOR CALCULATION

- ① Design a PDA to Accept strings with n occurrences of a followed n occurrences of b where n is non negative integer
 $L = \{a^n b^n\}$
- ② Convert the given CFG into a Pushdown Automata that recognize same degree

$$E \rightarrow E + E \mid E * E \mid (E) \mid 1$$

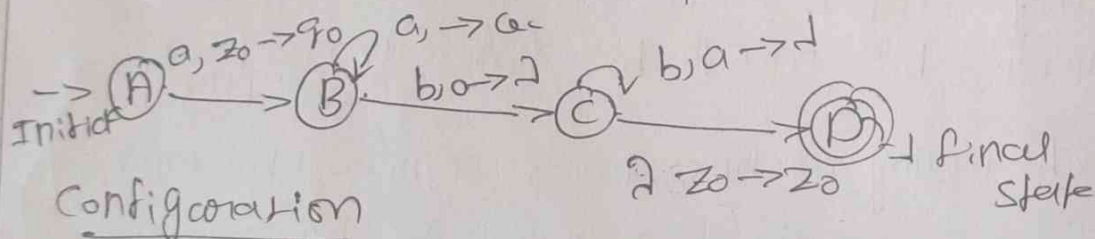
$$1 \rightarrow 10 \mid 110 \mid 1011 \mid 10110 \mid 101101$$
- ③ Design a Turing machine to accepts the string with n occurrences where n is non negative integer.
 $L = \{0^n 1^n\} (n \geq 1)$
- ④ using Pushdown automata for the language representing a of b 's must followed by b 's and the no. of b 's must be twice that of a 's
 $L = \{a^n b^{2n}\} \quad n \geq 1$
- ⑤ Construct Turing machine to recognize the language write the logic formal used for design and formal definition
 $L = \{a^n b^n c^n\}$

Solutions

- ① Given
 $L = \{a^n b^n\}$
 Let $n = 3 : L = a^3 b^3$

for every 'a'
 Push one 'A'
 for every 'b' POP one 'A'

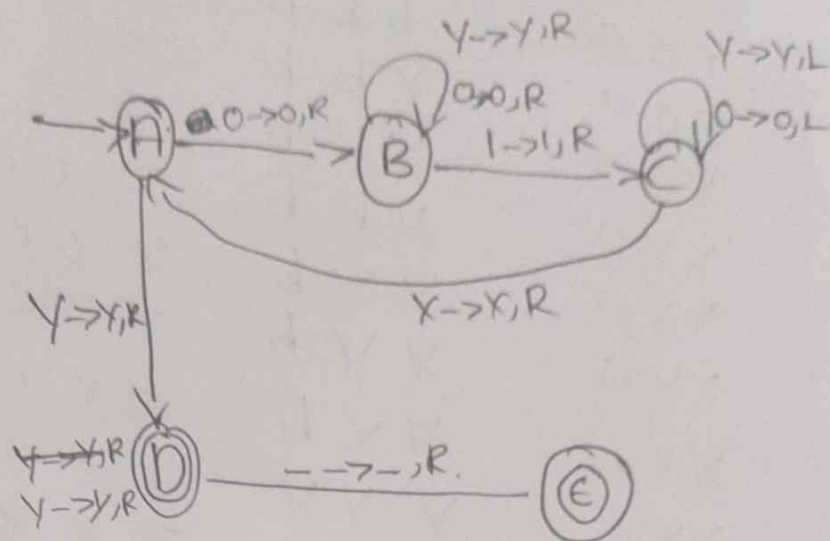
Stack	PDA	operation
aaa bbb	z_0	Push 'A'
aa bbb	A z_0	Push 'A'
a bbb	AA z_0	Push 'A'
bbb	AAA z_0	POP 'B'
bb	AA z_0	POP 'B'
b	A z_0	POP 'B'
ϵ	z_0	Accept
ϵ	z_0	



- ① $\delta(A, a, z_0) = (B, a, z_0)$
- ② $\delta(B, a, a) = \{B, a, a\}$
- ③ $\delta(B, b, a) = (C, \epsilon)$
- ④ $\delta(C, d, z_0) = (D, z_0)$
- ⑤ $\delta(C, b, a) = (C, \epsilon)$

given
 $L = \{0^n 1^n\} \quad n \geq 1$
 let $n=3$
 $L = \{0^3 1^3\}$

0	0	0	1	1	1
↓					
X	0	0	1	1	1
	↓		↓		
0	0	0	Y	1	1
↓			↓		
X	X	0	Y	1	1
	↓		↓		
X	X	0	Y	Y	1
	↓		↓		
X	X	X	Y	Y	1
			↓		
X	X	X	Y	Y	Y



Configurations

$$\delta(A, 0) = \{B, 0, R\}$$

$$\delta(B, 0) = \{B, 0, R\}$$

$$\delta(B, Y) = \{B, Y, R\}$$

$$\delta(B, 1) = \{C, 1, R\}$$

$$\delta(C, 0) = \{C, 0, L\}$$

$$\delta(C, Y) = \{C, Y, L\}$$

$$\delta(C, X) = \{A, X, R\}$$

$$\delta(A, Y) = \{D, Y, R\}$$

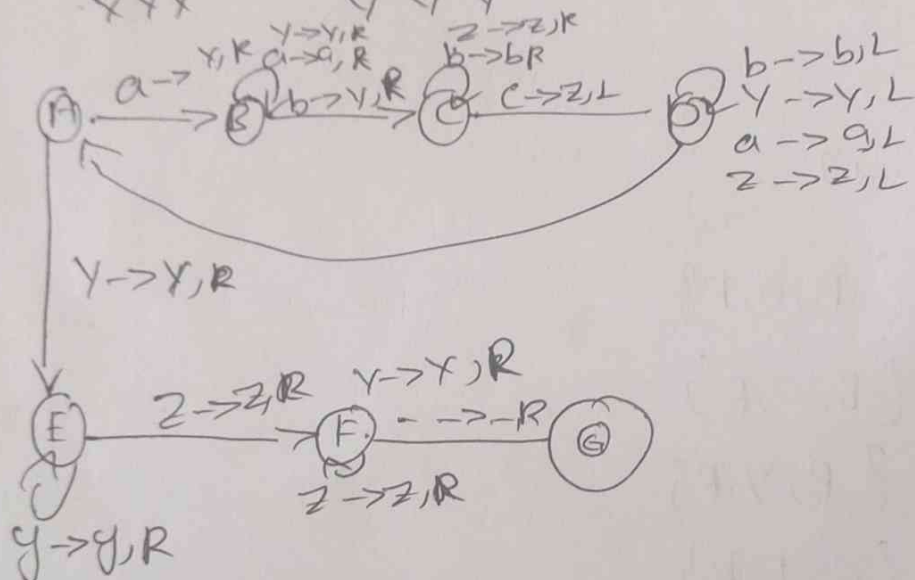
$$\delta(D, Y) = \{D, Y, R\}$$

$$\delta(D, -) = \{D, -, R\}$$

⑤ TM for $L = a^n b^n c^n$

$n=3$

aaa	bbb	ccc
xaa	bbb	ccc
yaa	ybb	ccc
yaa	ybb	zcc
xxa	ybb	zcc
xxa	yyb	zcc
xya	yyb	zzc
xyx	yyb	zzc
xyy	yyy	zzc
xyx	yyy	zzz



Configurations

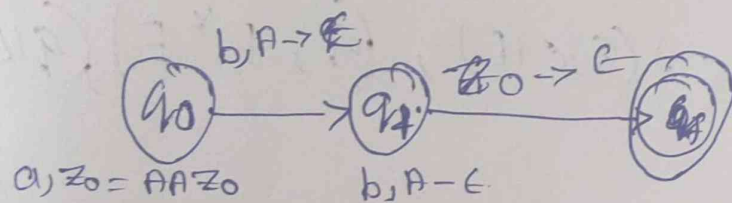
- ① $\delta(A, a) \rightarrow (B, x, R)$
- ② $\delta(B, a) \rightarrow (B, a, R)$
- ③ $\delta(B, b) \rightarrow (C, y, R)$
- ④ $\delta(C, b) \rightarrow (C, b, R)$
- ⑤ $\delta(C, c) \rightarrow (D, z, L)$
- ⑥ $\delta(D, b) \rightarrow (D, b, L)$
- ⑦ $\delta(D, y) \rightarrow (E, y, R)$
- ⑧ $\delta(D, a) \rightarrow (D, a, L)$
- ⑨ $\delta(D, z) \rightarrow (D, z, L)$
- ⑩ $\delta(D, x) \rightarrow (A, x, R)$
- ⑪ $\delta(A, y) \rightarrow (E, y, R)$
- ⑫ $\delta(F, y) \rightarrow (E, y, R)$
- ⑬ $\delta(F, z) \rightarrow (F, z, R)$
- ⑭ $\delta(F, -) \rightarrow (G, -, R)$
- ⑮ $\delta(F, -) \rightarrow (G, -, R)$
- ⑯ $\delta(B, y) \rightarrow (B, y, R)$
- ⑰ $\delta(C, z) \rightarrow (C, z, R)$

Design PDA for
 $L = \{a^n b^{2n} \mid n \geq 1\}$

so, abb, aabbbb, aaabbbbbbb $\rightarrow$ Accept
 ab, abaa, aabb $\rightarrow$ Reject
 Let $n=2$ so,
 aa bbbb

step 1) for every 'a' Push 2 'A's into stack
 step 2) for every 'b' Pop one 'A' from stack

$\delta(q_0, a, z_0) = (q_0, AAz_0)$
 $\delta(q_0, a, A) = (q_0, AAAz_0)$
 $\delta(q_1, b, A) = (q_1, \epsilon)$
 $\delta(q_1, b, A) = (q_1, \epsilon)$
 $\delta(q_1, b, A) = (q_1, \epsilon)$
 $\delta(q_1, b, A) = (q_1, \epsilon)$
 $\delta(q_1, z_0) = (q_f, \epsilon)$



Stack	Stack	Operation
aaabbbb	z0	Start
aaabbbb	AAz0	Push 'AA'
bbbbb	AAAAz0	Push 'AA'
bbbb	AAAz0	Pop 'AA'
bb	AAz0	Pop 'A'
b	Az0	Pop 'A'
ε	z0	Pop 'A'

② CFG into PDA

$E \rightarrow E + E \mid E * E \mid (E) \mid \epsilon$

$I \rightarrow a \mid b \mid \epsilon$

Here the language

$E \rightarrow E + E$	$I \rightarrow (a)$
$E \rightarrow E * E$	$I \rightarrow (b)$
$E \rightarrow (E)$	$I \rightarrow (a)$
$E \rightarrow \epsilon$	$I \rightarrow (b)$
	$I \rightarrow \epsilon$

$q = \{q_0, q_1, q_2\}$

$M = \{Q, \Sigma, q_0, q_f, \delta\}$

$\Sigma = \{a, b, +, *, \epsilon\}$

① Push the start symbol

$\delta(q_0, \epsilon, z_0) = \{q_1, (E, z_0)\}$

② Replace variables

for E , $\delta(q_1, E, E) = \{q_1, E + E\}, (q_1, E * E), (q_1, E), (q_1, \epsilon)\}$

for I , $\delta(q_1, I, \epsilon) = \{q_1, I, a\}, (q_1, I, b), (q_1, I, \epsilon), (q_1, I, a), (q_1, I, b), (q_1, a), (q_1, b)\}$

match terminals

$\delta(q_1, a, a) = \{q_1, \epsilon\}$

$\delta(q_1, b, b) = \{q_1, \epsilon\}$

$\delta(q_1, \epsilon, \epsilon) = \{q_1, \epsilon\}$

$\delta(q_1, +, +) = \{q_1, \epsilon\}$

$\delta(q_1, *, *) = \{q_1, \epsilon\}$

$\delta(q_1, \epsilon, \epsilon) = \{q_1, \epsilon\}$

$\delta(q_1, \epsilon) = \{q_1, \epsilon\}$

finally

$\delta(q_1, \epsilon, z_0) = \{q_1, z_0\}$